

Hypothesis Testing Summary

Business Question

Is there a significant relationship between Sales and Profit?

Null Hypothesis (H_0)

There is no significant relationship between Sales and Profit.

Alternative Hypothesis (H_a)

There is a significant relationship between Sales and Profit.

Statistical Method

Correlation Analysis (Pearson Correlation), 95% Confidence Level, Significance Level ($\alpha = 0.05$).

Procedure

1. Import the cleaned sales dataset.
2. Compare Sales and Profit columns.
3. Calculate the Pearson correlation coefficient.
4. Obtain the p-value.
5. Reject H_0 if p-value < 0.05 ; otherwise fail to reject H_0 .

Interpretation

A statistically significant positive correlation indicates that higher sales are associated with higher profit. Business decisions should be supported by statistical evidence together with dashboard insights.

Conclusion

The hypothesis testing framework validates business insights before making strategic decisions.